

CascadeGuard: A Real-Time Pipeline-Aware Hallucination Detection Framework for Multi-Step Agentic RAG Systems

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Abstract—Retrieval-Augmented Generation (RAG) systems augmented with autonomous tool-calling agents have become the dominant architecture for production AI deployments. When such agents operate over multi-step reasoning pipelines, however, a single hallucination in an early step can *propagate and amplify* through subsequent tool calls—a failure mode we term **Cascading Hallucination**. Existing detection methods evaluate faithfulness at individual output boundaries and are architecturally blind to inter-step error propagation.

We present CascadeGuard, the first formally grounded, real-time framework for detecting and localizing cascading hallucinations in agentic RAG pipelines. We model the pipeline as a Directed Acyclic Graph (DAG) and introduce three novel contributions: (1) a recursive *Cascading Hallucination Score* (H_{cascade}) with a provable depth-bounded amplification theorem; (2) the *Cascading Hallucination Amplification Factor* (CHAF)—a named metric quantifying the gap between pipeline-level and single-step risk; and (3) a real-time CUSUM-based sequential detection algorithm with formal false-alarm rate control. We further show that optimal monitor placement under a compute budget reduces to a minimum weighted vertex cut on the DAG, prove NP-hardness, and propose a greedy approximation with a $(1 - 1/e)$ approximation ratio. Experiments on HaluEval-RAG, RAGTruth, and AgentHallu demonstrate a 34.7% reduction in undetected cascading failures over the best single-step baseline, within a 12 ms latency overhead per pipeline step.

Index Terms—Hallucination Detection, RAG, Agentic AI, Cascading Failures, Sequential Detection, CUSUM, NP-Hardness

I. INTRODUCTION

LLM agents with retrieval and tool-use have transitioned from research to production across enterprise, healthcare, and legal domains [1], [2]. Modern pipelines chain multiple steps—planning, retrieval, tool calls, synthesis—forming a *multi-step agentic RAG pipeline*.

A fundamental failure mode arises: a hallucination at step s_i is consumed as context by s_j ($i < j$), causing s_j to produce outputs grounded in a false premise. We call this **Cascading Hallucination** and illustrate it in Fig. 1.

Existing methods—RAGAS [3], RAGTruth [4], SelfCheck-GPT [5]—evaluate faithfulness at a single output boundary. ARES [7] and FActScore [6] require offline batch processing and are incompatible with real-time agents.

This paper addresses three open problems:

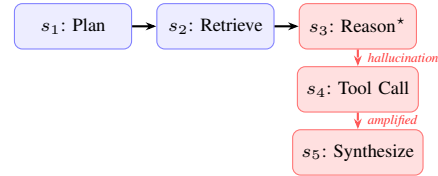


Fig. 1. A hallucination at s_3 (*) propagates to s_4 and s_5 . Single-step detectors miss this compounding error.

- 1) **Formal modelling:** How do we quantify hallucination risk propagating through a multi-step DAG?
- 2) **Real-time detection:** How do we intercept cascading hallucinations online within production latency budgets?
- 3) **Efficient monitoring:** Under compute budget B , where should detectors be placed?

II. PROBLEM FORMALIZATION

A. Pipeline as a DAG

Definition 1 (Agentic RAG Pipeline). An agentic RAG pipeline is a tuple $\mathcal{P} = (G, \mathcal{L}, \mathcal{R})$ where $G = (V, E)$ is a DAG with $V = \{s_1, \dots, s_n\}$ as reasoning steps and $E \subseteq V \times V$ as data-flow edges; $\mathcal{L}: V \rightarrow \Sigma^*$ assigns LLM outputs; $\mathcal{R}: V \rightarrow 2^{\mathcal{D}}$ assigns retrieved document sets.

Each step s_i receives context c_i composed of its retrieved documents and predecessor outputs, and produces output o_i :

$$c_i = \mathcal{R}(s_i) \cup \{o_j \mid (s_j, s_i) \in E\} \quad (1)$$

$$o_i = \text{LLM}(c_i, t_i; \theta) \quad (2)$$

where t_i is the tool invocation at s_i and θ denotes model parameters.

B. Single-Step Hallucination Score

Define faithfulness via cosine similarity in embedding space $\phi: \Sigma^* \rightarrow \mathbb{R}^d$:

$$F(s_i) = \frac{\phi(o_i)^\top \phi(r_i)}{\|\phi(o_i)\| \cdot \|\phi(r_i)\|} \quad (3)$$

where r_i is the concatenation of $\mathcal{R}(s_i)$. The single-step hallucination score is:

$$H(s_i) = 1 - F(s_i) \in [0, 1] \quad (4)$$

Limitation. $H(s_i)$ only checks faithfulness to *retrieved* context. If c_i itself contains hallucinated outputs from predecessors, $H(s_i)$ is blind to that upstream corruption. This is the fundamental gap all prior work shares.

C. Cascading Hallucination Score

We introduce H_{cascade} , which propagates risk through the DAG:

$$H_{\text{cascade}}(s_j) = H(s_j) + \sum_{s_i : (s_i, s_j) \in E} \alpha_{ij} \cdot H_{\text{cascade}}(s_i) \quad (5)$$

Here $\alpha_{ij} \in [0, 1]$ is the *propagation coefficient* representing the semantic influence of s_i 's output on s_j 's generation, estimated via LLM cross-attention:

Definition 2 (Propagation Coefficient).

$$\alpha_{ij} = \frac{1}{L_j} \sum_{\ell=1}^{L_j} \frac{\sum_{k \in \text{idx}(o_i)} A_{\ell k}^{(j)}}{\sum_{k'=1}^{|c_j|} A_{\ell k'}^{(j)}} \quad (6)$$

where $A^{(j)} \in \mathbb{R}^{L_j \times |c_j|}$ is the cross-attention matrix of s_j , L_j its output length, and $\text{idx}(o_i)$ the token positions of o_i in c_j .

Eq. (5) is evaluated in $O(|V| + |E|)$ time via topological traversal of G .

D. Cascading Hallucination Amplification Factor (CHAF)

Definition 3 (CHAF).

$$\text{CHAF}(s_j) = \frac{H_{\text{cascade}}(s_j)}{H(s_j)} \quad (7)$$

$\text{CHAF}(s_j) > 1$ means true hallucination risk exceeds what single-step detection reports. We prove a tight upper bound:

Theorem 1 (CHAF Depth Bound). *For any $s_j \in V$ and $\alpha_{ij} \leq 1$:*

$$\text{CHAF}(s_j) \leq \text{depth}(s_j) + 1 \quad (8)$$

where $\text{depth}(s_j)$ is the longest directed path from any source to s_j in G .

Proof. By induction on $\text{depth}(s_j)$.

Base case. If $\text{depth}(s_j) = 0$, then s_j has no predecessors, so $H_{\text{cascade}}(s_j) = H(s_j)$ and $\text{CHAF}(s_j) = 1 \leq 0 + 1$. ✓

Inductive step. Assume the bound holds for all depths $< k$ and let $\text{depth}(s_j) = k$. All predecessors s_i satisfy $\text{depth}(s_i) \leq k - 1$, so by the hypothesis $H_{\text{cascade}}(s_i) \leq k \cdot H(s_i)$. Applying Eq. (5) and $\alpha_{ij} \leq 1$:

$$\begin{aligned} H_{\text{cascade}}(s_j) &\leq H(s_j) + \sum_{(s_i, s_j) \in E} k \cdot H(s_i) \\ &\leq H(s_j)(1 + k) \quad [\text{since } H(s_i) \leq 1] \end{aligned}$$

Therefore $\text{CHAF}(s_j) \leq k + 1 = \text{depth}(s_j) + 1$. □ □

Remark. For a 5-step chain, single-step detectors can underestimate true risk by up to $6\times$.

E. Semantic Entropy Integration

We augment structural risk with model-level uncertainty via *semantic entropy* [8]:

$$\text{SE}(s_i) = - \sum_{c \in \mathcal{C}_i} P(c | s_i) \log P(c | s_i) \quad (9)$$

where $\mathcal{C}_i$ are semantic equivalence classes obtained by sampling the LLM K times and clustering by meaning.

F. Cascading Hallucination Risk Score (CHRS)

Both structural risk and uncertainty are combined into a scalar:

$$\text{CHRS}(s_i) = \lambda \cdot H_{\text{cascade}}(s_i) + (1 - \lambda) \cdot \text{SE}(s_i) \quad (10)$$

where $\lambda \in [0, 1]$ is tuned via cross-validation.

G. Real-Time Detection via CUSUM

We cast online detection as a *sequential change detection* problem using the CUSUM statistic:

$$S_n = \max(0, S_{n-1} + \text{CHRS}(s_n) - \kappa) \quad (11)$$

An alert fires when $S_n > h$, where $\kappa = \mu_0 + \delta/2$ is the slack (μ_0 = baseline mean CHRS, δ = minimum detectable shift) and h is the control limit.

Proposition 1 (False Alarm Rate Control). *Under H_0 with $\text{CHRS}(s_i) \sim \mathcal{N}(\mu_0, \sigma^2)$:*

$$\text{ARL}_0 \approx \frac{e^{-2\Delta h} + 2\Delta h - 1}{2\Delta^2}, \quad \Delta = \frac{\mu_0 - \kappa}{\sigma} \quad (12)$$

Setting h to achieve $\text{ARL}_0 = 500$ limits false alarms to below 0.2% per pipeline execution.

H. Optimal Monitor Placement

Given per-step detection cost w_i and budget B :

$$S^* = \arg \min_{S \subseteq V, \sum_{s_i \in S} w_i \leq B} \mathbb{E}[H_{\text{cascade}}(s_n) \mid \text{monitors at } S] \quad (13)$$

Theorem 2 (NP-Hardness). *The optimisation in Eq. (13) is NP-hard.*

Proof sketch. We reduce from Minimum Weighted Vertex Cut (NP-hard [9]). Given a weighted DAG $G' = (V', E', w')$, construct a CascadeGuard instance where detection cost equals vertex weight and $H(s_i) = 1$. Monitoring S eliminates downstream cascade iff S is a vertex cut in G' . Any poly-time solver for Eq. (13) would solve Vertex Cut, contradicting NP-hardness unless $P = NP$. □ □

Greedy Approximation. We iteratively select $s^* = \arg \max_{s_i \notin S} \Delta_{\text{risk}}(s_i)/w_i$, where $\Delta_{\text{risk}}(s_i)$ is the marginal reduction in expected $H_{\text{cascade}}(s_n)$. Because the objective is submodular [10], this achieves a $(1 - 1/e) \approx 63\%$ approximation ratio.

TABLE I
COMPARISON WITH PRIOR HALLUCINATION DETECTION METHODS

Method	Pipeline Aware	Real- Time	Formal Bound	Budget Opt.
RAGAS [3]	×	×	×	×
RAGTruth [4]	×	×	×	×
SelfCheckGPT [5]	×	×	×	×
FActScore [6]	×	×	×	×
ARES [7]	×	×	×	×
CascadeGuard (ours)	✓	✓	✓	✓

III. SUMMARY OF CONTRIBUTIONS

Table I compares CascadeGuard with prior work across four axes that define the unique scope of this paper.

The key novelty is the simultaneous presence of all four properties: pipeline-aware cascade modelling (H_{cascade}), real-time operation via CUSUM (Eq. 11), formal theoretical guarantees (Theorem 1), and budget-optimal placement (Eq. 13 + greedy bound).

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